



# Jaume Ivars Grimalt

MACHINE LEARNING ENGINEER · COMPUTER VISION & PRODUCTION SYSTEMS

Valencia, Spain — remote across EU/US timezones

☎ (+34) 690183644 | ✉ jaume@jaumeivars.com | 🌐 jaumeivars.com | 📀 jaume2000

## Profile

I build production computer-vision systems end to end: mathematical formulation, training, deployment, and the product people use. At Mycrospace, formulated high-density colony counting as density estimation, designed proprietary Density Map Regressors with Bayesian priors (**67 mAP50-95**), and shipped a product used by **5 laboratories** in production. I train models at scale on European HPC and apply that mathematical base in domains where reliability matters — microbiology and medical imaging.

## Technical Skills

### Computer Vision

Detection, segmentation, density estimation

### ML Engineering

Training at scale on HPC, PyTorch, TensorFlow, Production ML (Docker, GCP, AWS, CI/CD)

### Backend

NestJS, Prisma, PostgreSQL, MongoDB

### Frontend

Next.js, TypeScript

## Experience

### Mycrospace

Valencia, Spain

CO-FOUNDER & CTO

Jan 2024 – Present

Computer-vision product for microbiology, in production. Concurrently part-time with MIALAB (May 2025 – Jan 2026).

- Formulated high-density colony counting (>300 overlapping colonies per plate) as density estimation; designed proprietary Density Map Regressors with Bayesian priors. Object detection at **67 mAP50-95** and **95% median accuracy** against expert microbiologist annotation.
- Built the full system: PyTorch training framework, data pipelines, containerized inference (Docker/GCP), REST API (NestJS/Prisma), webapp (Next.js/TypeScript). Own dataset of **3,500** annotated images across **12** microbial species; annotation protocols co-designed with microbiologists. **3,012 GPU-hours** of training.
- **5 laboratories** using the product in production today. Supervise one full-stack developer (frontend and backend) since **[TODO: since when]**.

### MIALAB — Medical Imaging Analysis Laboratory, UPV

Valencia, Spain

ML ENGINEER, PART-TIME

May 2025 – Jan 2026

- nnU-Net models for volumetric brain MRI segmentation, with distinct multi-scale feature-extraction strategies.
- Inference-pipeline optimization for the **VolBrain** platform (**700k+** MRI volumes): bottleneck analysis from NIfTI ingest to voxel-level prediction.

### Neurocatching

Valencia, Spain

ML ENGINEER

2022 – 2023

- Temporal gaze-pattern models from raw ocular-movement sequences.

### Sciling

Valencia, Spain

AI CONSULTANT

2022

- AI consulting for external clients: conversational systems on GPT-3-era LLMs, applied NLP, and generative models.
- **[TODO: what I built, exactly, at Sciling]**

**[TODO: N]** h/week, remote or hybrid from Valencia, alongside my role as CTO of Mycrospace.

## Projects

---

### ConvNeXt-V1 trained from scratch

IMAGENET-1K REPRODUCTION ON EUROHPC (LEONARDO)

- **1,687 GPU-hours** consumed to date; training ongoing. [TODO: final top-1]. [TODO: GPU count and type]. [TODO: technical problem diagnosed]. [github.com/jaume2000/convnext\\_v1](https://github.com/jaume2000/convnext_v1)

### Universal two-tape Turing machine

TYPESCRIPT + CANVAS, RUNS IN THE BROWSER

[github.com/jaume2000/lvarsTurinMachine](https://github.com/jaume2000/lvarsTurinMachine)

### CanvasGraph

INTERACTIVE CANVAS ANIMATIONS

[github.com/jaume2000/CanvasGraph](https://github.com/jaume2000/CanvasGraph)

## Education

---

### Universitat Politècnica de València (UPV)

M.SC. IN ARTIFICIAL INTELLIGENCE, COMPUTER VISION & DIGITAL IMAGE

Master's Thesis with Honours ([TODO: thesis year]).

*Valencia, Spain*

Sep 2022 – [TODO: master end]

### Universitat Politècnica de València (UPV)

B.SC. IN COMPUTER SCIENCE

Honor Mentions in Machine Learning and Statistics ([TODO: mentions year]).

*Valencia, Spain*

Sep 2017 – Jun 2022

## Awards

---

2024    **1st Prize, IdeasUPV Startup Competition**, Mycrospace — €10,000 grant

*Valencia, Spain*